SVM_Day04

March 6, 2026

[29]:
```
import pandas as pd
import numpy as np
```

[30]:
```
data=pd.read_csv('Iris.csv')
```

[31]:
```
data.sample(5)
```

[31]:
```
      Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  \
129  130            7.2           3.0            5.8           1.6
45    46            4.8           3.0            1.4           0.3
127  128            6.1           3.0            4.9           1.8
144  145            6.7           3.3            5.7           2.5
61    62            5.9           3.0            4.2           1.5

             Species
129   Iris-virginica
45       Iris-setosa
127   Iris-virginica
144   Iris-virginica
61   Iris-versicolor
```

[32]:
```
data=data.drop('Id',axis=1)
```

[33]:
```
data.head()
```

[33]:
```
   SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0            5.1           3.5            1.4           0.2  Iris-setosa
1            4.9           3.0            1.4           0.2  Iris-setosa
2            4.7           3.2            1.3           0.2  Iris-setosa
3            4.6           3.1            1.5           0.2  Iris-setosa
4            5.0           3.6            1.4           0.2  Iris-setosa
```

[34]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
```

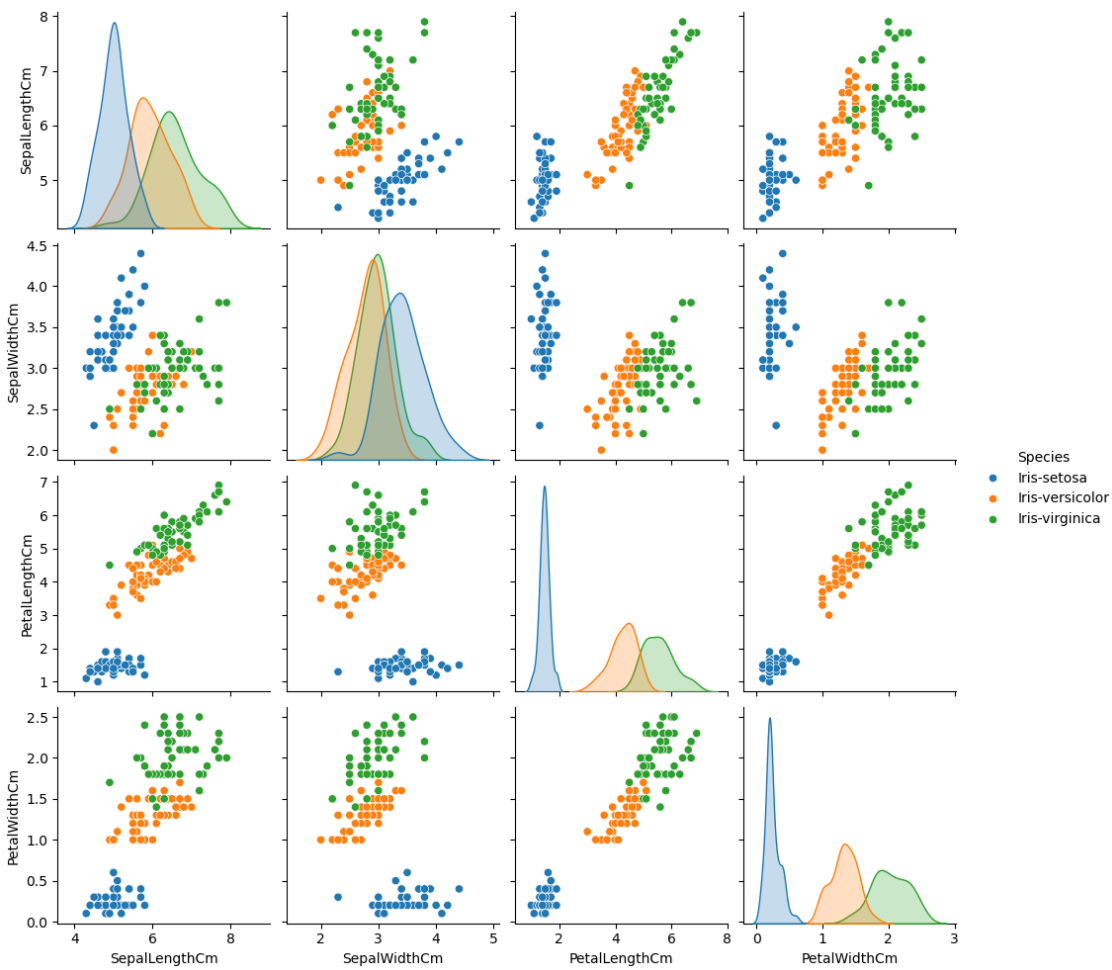
```
 0   SepalLengthCm  150 non-null    float64
 1   SepalWidthCm   150 non-null    float64
 2   PetalLengthCm  150 non-null    float64
 3   PetalWidthCm   150 non-null    float64
 4   Species        150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

[35]:
```
data['Species'].value_counts()
```

[35]:
```
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

[36]:
```
import seaborn as sns
sns.pairplot(data, hue='Species')
```

[36]:
```
<seaborn.axisgrid.PairGrid at 0x21c4b2e5e50>
```



[37]:
```
x = data.drop('Species', axis=1)
x.sample(5)
```

[37]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
76	6.8	2.8	4.8	1.4
114	5.8	2.8	5.1	2.4
80	5.5	2.4	3.8	1.1
70	5.9	3.2	4.8	1.8
30	4.8	3.1	1.6	0.2

[38]:
```
y=data['Species']
y.sample(5)
```

[38]:
```
12         Iris-setosa
26         Iris-setosa
49         Iris-setosa
89     Iris-versicolor
122     Iris-virginica
Name: Species, dtype: object
```

[39]:
```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)
```

[40]:
```
from sklearn.svm import SVC
model = SVC(kernel='poly',C=10)
model.fit(x_train, y_train)
```

[40]:
```
SVC(C=10, kernel='poly')
```

[41]:
```
model.kernel
```

[41]:
```
'poly'
```

[42]:
```
pred = model.predict(x_test)
```

[43]:
```
pred
```

[43]:
```
array(['Iris-versicolor', 'Iris-versicolor', 'Iris-versicolor',
       'Iris-versicolor', 'Iris-setosa', 'Iris-versicolor',
       'Iris-virginica', 'Iris-virginica', 'Iris-virginica',
       'Iris-setosa', 'Iris-setosa', 'Iris-virginica', 'Iris-setosa',
       'Iris-virginica', 'Iris-virginica', 'Iris-versicolor',
       'Iris-versicolor', 'Iris-versicolor', 'Iris-setosa',
       'Iris-virginica', 'Iris-versicolor', 'Iris-virginica',
       'Iris-versicolor', 'Iris-versicolor', 'Iris-virginica',
```

```
       'Iris-virginica', 'Iris-versicolor', 'Iris-versicolor',
       'Iris-setosa', 'Iris-virginica'], dtype=object)
```

[44]:
```
y_test
```

[44]:
```
88     Iris-versicolor
52     Iris-versicolor
50     Iris-versicolor
95     Iris-versicolor
2          Iris-setosa
91     Iris-versicolor
110     Iris-virginica
146     Iris-virginica
122     Iris-virginica
25         Iris-setosa
15         Iris-setosa
106     Iris-virginica
22         Iris-setosa
100     Iris-virginica
139     Iris-virginica
56     Iris-versicolor
63     Iris-versicolor
97     Iris-versicolor
42         Iris-setosa
125     Iris-virginica
84     Iris-versicolor
70     Iris-versicolor
80     Iris-versicolor
92     Iris-versicolor
124     Iris-virginica
123     Iris-virginica
90     Iris-versicolor
76     Iris-versicolor
5          Iris-setosa
105     Iris-virginica
Name: Species, dtype: object
```

[45]:
```
from sklearn.metrics import accuracy_score,confusion_matrix
```

[46]:
```
accuracy_score(pred,y_test)
```

[46]:
```
0.9666666666666667
```

[47]:
```
confusion_matrix(pred,y_test)
```

[47]:
```
array([[ 6,  0,  0],
       [ 0, 13,  0],
```

```
       [ 0,  1, 10]])
```

[ ]:

```

```